

Next, observe that for $\alpha \in (0, 1)$ the probability in (3.18) can be bounded as follows:

$$\begin{aligned}
& \mathbb{P} \left(\max_{k \leq m} \left(\frac{k+1}{m} \right)^\gamma |k^{-1} \bar{Z}_k - \mathbb{E}[Z_0]| > \varepsilon \right) \\
& \leq \mathbb{P} \left(\max_{k \leq m} \left(\frac{k+1}{m} \right)^\gamma |k^{-1} \bar{Z}_k - \mathbb{E}[Z_0]| > \varepsilon, M_0 \leq M_1 \right) + \mathbb{P}(M_0 > M_1) \\
& \leq \mathbb{P} \left(\max_{k \leq \lfloor m^\alpha \rfloor} \left(\frac{k+1}{m} \right)^\gamma |k^{-1} \bar{Z}_k - \mathbb{E}[Z_0]| > \varepsilon, M_0 \leq M_1 \right) \\
& \quad + \mathbb{P} \left(\max_{\lfloor m^\alpha \rfloor < k \leq m} \left(\frac{k+1}{m} \right)^\gamma |k^{-1} \bar{Z}_k - \mathbb{E}[Z_0]| > \varepsilon, M_0 \leq M_1 \right) + \frac{\delta}{2} \\
& \leq \mathbb{P} \left(\left(\frac{\lfloor m^\alpha \rfloor + 1}{m} \right)^\gamma \max_{k \leq \lfloor m^\alpha \rfloor} |k^{-1} \bar{Z}_k - \mathbb{E}[Z_0]| > \varepsilon \right) \\
& \quad + \mathbb{P} \left(\max_{\lfloor m^\alpha \rfloor < k \leq m} |k^{-1} \bar{Z}_k - \mathbb{E}[Z_0]| > \frac{\varepsilon}{2^\gamma}, M_0 \leq M_1 \right) + \frac{\delta}{2} \tag{3.20}
\end{aligned}$$

Furthermore, applying Markov's inequality to the first term in (3.20) shows

$$\begin{aligned}
& \mathbb{P} \left(\left(\frac{\lfloor m^\alpha \rfloor + 1}{m} \right)^\gamma \max_{k \leq \lfloor m^\alpha \rfloor} |k^{-1} \bar{Z}_k - \mathbb{E}[Z_0]| > \varepsilon \right) \\
& \leq \frac{1}{\varepsilon} \left(\frac{\lfloor m^\alpha \rfloor + 1}{m} \right)^\gamma \mathbb{E} \left[\max_{k \leq \lfloor m^\alpha \rfloor} |k^{-1} \bar{Z}_k - \mathbb{E}[Z_0]| \right] \\
& \leq \frac{1}{\varepsilon} \left(m^{\alpha-1} + \frac{1}{m} \right)^\gamma m^\alpha \mathbb{E}[|Z_1 - \mathbb{E}[Z_0]|] \\
& \leq \frac{1}{\varepsilon} \left(m^{\alpha(\gamma+1)-\gamma} + m^{\alpha-\gamma} \right) \mathbb{E}[|Z_1 - \mathbb{E}[Z_0]|]. \tag{3.21}
\end{aligned}$$

Hence, if we let $0 < \alpha < \gamma/(1+\gamma)$ both exponents of m become negative and we can find $M_2 \in \mathbb{N}$ such that (3.21) is less than $\delta/2$ for all $m \geq M_2$. Finally, we put $M := M_1^{1/\alpha} \vee M_2$. Then $m \geq M$ implies $m^\alpha \geq M_1$ and the second term in (3.20) vanishes therefore for all $m \geq M$ due to (3.19), which completes the proof. $\square$

Corollary 3.3. *Any stationary, L^2 - m -approximable sequence $(Z_n)_n$ satisfies the strong law of large numbers, that is $n^{-1}(Z_1 + \dots + Z_n) \rightarrow \mathbb{E}[Z_0]$ almost surely.*

Proof. Without loss of generality we can assume $\mathbb{E}[Z_0] = 0$ by centering. First observe that Z_0 and $Z_j^{(j)}$ are independent and therefore

$$\begin{aligned}
\sum_{j \in \mathbb{N}} |\mathbb{E}[Z_0 Z_j]| & \leq \sum_{j \in \mathbb{N}} |\mathbb{E}[Z_0(Z_j - Z_j^{(j)})]| + \sum_{j \in \mathbb{N}} |\mathbb{E}[Z_0 Z_j^{(j)}]| \\
& \leq \mathbb{E}[Z_0^2]^{1/2} \sum_j \mathbb{E}[(Z_0 - Z_0^{(j)})^2]^{1/2} < \infty.
\end{aligned}$$

In particular for any $I \subseteq \mathbb{N}$, $\text{Var}(\sum_{j \in I} Z_j) \leq C \#I$ for a constant $C \in \mathbb{R}_+$ which does not depend on I . Consequently, for $S_n = \sum_{i=1}^n Z_i$, we can rely on a classical proof technique for the strong law of large numbers and use the Borel-Cantelli lemma to see that both

$$\frac{1}{n^2} S_{n^2} \rightarrow 0 \quad a.s. \quad \text{and} \quad \max_{n^2+1 \leq k \leq (n+1)^2} \frac{1}{n^2} (S_k - S_{n^2}) \rightarrow 0 \quad a.s.$$

In particular, $\max_{n^2+1 \leq k \leq (n+1)^2} \frac{1}{k} S_k \rightarrow 0$ for $n \rightarrow \infty$ with probability 1. Clearly, this implies the strong law of large numbers. $\square$

3.2. Two sample tests for inco-variances

Lemma 3.4 (Cohen-Steiner et al. [11]). *Let $1 \leq p' \leq r < \infty$ and let $\text{PD}(\mathcal{X}), \text{PD}(\mathcal{Y})$ be persistence diagrams whose p' -total persistence is bounded from above. Then*

$$\begin{aligned} W_r(D, E) &\leq \left(\frac{\text{Pers}_{p'}(\text{PD}(\mathcal{X})) + \text{Pers}_{p'}(\text{PD}(\mathcal{Y}))}{2} \right)^{1/r} W_\infty(\text{PD}(\mathcal{X}), \text{PD}(\mathcal{Y}))^{1-p'/r} \\ &\leq \left(\frac{\text{Pers}_{p'}(D) + \text{Pers}_{p'}(E)}{2} \right)^{1/r} d_H(\mathcal{X}, \mathcal{Y})^{1-p'/r}. \end{aligned}$$

Here the p' -total persistence is defined as $\text{Pers}_{p'}(D) = \sum_{(d_x, b_x) \in D} (d_x - b_x)^{p'}$.

Lemma 3.5 (Kusano et al. [22][Corollary 5]). *Let $\mathfrak{D}$ be a compact triangulable subset of $\mathbb{R}^d$. Let $\mathcal{X}$ and $\mathcal{Y}$ be finite subsets of $\mathfrak{D}$ and let $r \geq p' > d$. Then we have for the persistence diagrams $\text{PD}(\mathcal{X})$ and $\text{PD}(\mathcal{Y})$ from the Čech filtration (for any dimension)*

$$\begin{aligned} W_r(\text{PD}(\mathcal{X}), \text{PD}(\mathcal{Y})) &\leq \left(\frac{p'}{p' - d} C_{\mathfrak{D}} (\text{diam } \mathfrak{D})^{p' - d} \right)^{1/r} W_\infty(\text{PD}(\mathcal{X}), \text{PD}(\mathcal{Y}))^{1-p'/r} \\ &\leq \left(\frac{p'}{p' - d} C_{\mathfrak{D}} (\text{diam } \mathfrak{D})^{p' - d} \right)^{1/r} d_H(\mathcal{X}, \mathcal{Y})^{1-p'/r} \end{aligned}$$

for a constant $C_{\mathfrak{D}} \in \mathbb{R}_+$ which only depends on $\mathfrak{D}$.

Lemma 3.6.

(A) Let $d < r$.

(1) The kernel (1.13) is ρ -Hölder continuous w.r.t. the product Hausdorff metric on $\mathbb{N} \times \mathbb{N}$ for each $\rho \in (0, 1 - d/r)$

$$|h(\mathcal{X}, \mathcal{Y}) - h(\mathcal{X}', \mathcal{Y}')| \leq L_\rho (d_H(\mathcal{X}, \mathcal{X}') + d_H(\mathcal{Y}, \mathcal{Y}'))^\rho,$$

where the Hölder constant satisfies

$$L_\rho = 4 \left(\frac{r}{r - d} \frac{r(1 - \rho)}{r(1 - \rho) - d} C_{\mathfrak{D}}^2 (\text{diam } \mathfrak{D})^{r(2 - \rho) - 2d} \right)^{1/r}.$$

- (2) h satisfies for any $\mathcal{U}, \mathcal{V} \in \mathcal{N}|_{\mathfrak{D}}$ (i.e., point processes which are restricted to $\mathfrak{D}$)

$$h(\mathcal{U}, \mathcal{V}) \leq \left(\frac{r}{r-d} C_{\mathfrak{D}} (\text{diam } \mathfrak{D})^{r-d} \right)^{1/r} =: C^*.$$

- (B) Let $\mathcal{X}, \mathcal{Y}, \mathcal{X}', \mathcal{Y}'$ be finite with cardinality at most T .

- (1) h satisfies for any $\mathcal{X}, \mathcal{Y} \in \mathcal{N}|_{\mathfrak{D}}$ with cardinality at most T

$$h(\mathcal{X}, \mathcal{Y}) \leq \left((T^{k+1} + T^{k+2}) (\text{diam } \mathfrak{D})^r \right)^{1/r} =: C_*$$

if the underlying dimension of the persistence diagram is k .

- (2) The kernel (1.13) is $1 - 1/r$ -Hölder continuous w.r.t. the product Hausdorff metric on $\mathcal{N} \times \mathcal{N}$.

Proof. Part (A). We begin with statement (ii) which is a direct consequence of Lemma 3.5 in the special case $p' = r$. Note that both $\mathcal{U}|_{\mathfrak{D}}$ and $\mathcal{V}|_{\mathfrak{D}}$ are finite with probability 1 because $\mathfrak{D}$ is compact and elements of $\mathcal{N}$ are locally finite.

For statement (i), we apply the reverse triangle inequality

$$\begin{aligned} |h(\mathcal{X}, \mathcal{Y}) - h(\mathcal{X}', \mathcal{Y}')| &= |W_r^2(\text{PD}(\mathcal{X}), \text{PD}(\mathcal{Y})) - W_r^2(\text{PD}(\mathcal{X}'), \text{PD}(\mathcal{Y}'))| \quad (3.22) \\ &\leq \left\{ W_r(\text{PD}(\mathcal{X}), \text{PD}(\mathcal{Y})) + W_r(\text{PD}(\mathcal{X}'), \text{PD}(\mathcal{Y}')) \right\} \\ &\quad \cdot \left\{ W_r(\text{PD}(\mathcal{X}), \text{PD}(\mathcal{X}')) + W_r(\text{PD}(\mathcal{Y}), \text{PD}(\mathcal{Y}')) \right\}. \quad (3.23) \end{aligned}$$

Again using Lemma 3.5, the first factor on the RHS is uniformly bounded above as follows

$$W_r(\text{PD}(\mathcal{X}), \text{PD}(\mathcal{Y})) + W_r(\text{PD}(\mathcal{X}'), \text{PD}(\mathcal{Y}')) \leq 2C^*.$$

Moreover, for each $p' \in (d, r)$ the second factor is at most

$$\begin{aligned} &\left(\frac{p'}{p'-d} C_{\mathfrak{D}} (\text{diam } \mathfrak{D})^{p'-d} \right)^{1/r} \left\{ d_H(\mathcal{X}, \mathcal{X}')^{1-p'/r} + d_H(\mathcal{Y}, \mathcal{Y}')^{1-p'/r} \right\} \\ &\leq 2 \left(\frac{p'}{p'-d} C_{\mathfrak{D}} (\text{diam } \mathfrak{D})^{p'-d} \right)^{1/r} \left\{ d_H(\mathcal{X}, \mathcal{X}') + d_H(\mathcal{Y}, \mathcal{Y}') \right\}^{1-p'/r}. \end{aligned}$$

Hence, the kernel h is Hölder continuous for each $\rho \in (0, 1 - d/r)$ with Hölder constant

$$4C^* \left(\frac{r(1-\rho)}{r(1-\rho)-d} C_{\mathfrak{D}} (\text{diam } \mathfrak{D})^{r(1-\rho)-d} \right)^{1/r}.$$

Part (B). We rely this time on Lemma 3.4 and the trivial upper bound $\text{Pers}_{p'}(\text{PD}(\mathcal{X})) \leq (T^{q+1} + T^{q+2})(\text{diam } \mathfrak{D})^{p'}$, the calculations are very similar and we skip them for sake of brevity. $\square$